

PETROZAVODSK, Russian Federation

WMO: 228200

Lat: 61.8164N

Lon: 34.2550E

Elev: 111

StdP: 100.00

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-23.9	-29.6	0.2	-27.1	-26.0	0.4	-23.8	8.8	-2.8	8.0	-5.8	1.3	240	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	26.7	19.2	24.7	18.1	22.9	17.0	20.3	25.0	19.0	23.1	17.9	21.6	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	23.0	17.4	12.6	21.2	16.3	11.7	20.0	58.9	25.1	54.5	23.2	51.0	21.7	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.5	6.6	5.9		-29.7	29.2	3.9	2.3	-32.4	30.8	-34.7	32.1	-36.9	33.4	-39.7	35.1	
(5)			DB	-29.8	21.7	3.8	1.6	-32.6	22.9	-34.8	23.9	-36.9	24.8	-39.7	26.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-8.9	-8.9	-3.9	2.7	8.8	13.9	17.2	15.1	10.2	4.0	-1.9	-5.8
	DBStd	10.38	7.40	7.47	5.21	4.43	4.82	4.09	3.51	3.41	3.54	4.00	4.85	6.54
	HDD10.0	2939	586	529	431	224	81	9	1	2	40	189	357	490
	HDD18.3	5422	844	762	689	471	295	143	64	110	244	444	607	748
	CDD10.0	608	0	0	0	4	45	127	223	161	46	3	0	0
	CDD18.3	48	0	0	0	0	1	11	27	10	0	0	0	0
	CDH23.3	322	0	0	0	0	17	78	168	58	1	0	0	0
	CDH26.7	53	0	0	0	0	4	13	28	8	0	0	0	0
(14) Wind	WSAvg	2.8	2.8	2.9	3.1	3.0	3.0	2.7	2.3	2.4	2.5	2.9	3.0	3.1
(15) Precipitation	PrecAvg	574	30	22	30	35	42	59	69	81	68	58	42	39
	PrecMax	785	52	42	69	100	94	120	169	187	188	124	89	94
	PrecMin	380	3	5	10	5	2	12	4	33	16	0	6	13
	PrecStd	97	11	9	13	20	18	30	35	37	37	25	18	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	3.6	10.0	18.0	25.6	28.3	29.5	27.8	21.7	14.6	8.2	6.6
		MCWB	2.1	2.5	4.7	10.7	14.9	19.4	20.5	20.5	16.1	11.7	7.2	5.6
	2%	DB	1.7	2.2	6.1	14.4	21.7	25.4	27.0	24.9	18.8	12.1	7.0	4.0
		MCWB	1.1	1.5	2.4	8.6	13.5	18.1	19.9	18.6	14.6	10.4	6.1	3.3
	5%	DB	1.0	1.3	3.9	11.9	19.0	23.3	25.2	22.9	16.9	10.7	5.8	2.6
		MCWB	0.5	0.8	1.5	7.1	12.1	16.6	18.9	17.3	13.8	9.4	5.1	2.1
	10%	DB	0.1	0.5	2.3	9.2	16.9	21.1	23.2	20.9	15.2	9.6	4.5	1.5
		MCWB	-0.3	-0.1	0.7	5.2	11.0	15.4	17.7	16.3	12.8	8.4	3.9	1.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.7	4.9	11.6	17.1	21.1	22.3	21.3	17.0	12.3	7.4	5.6
		MCDB	2.4	3.2	9.2	17.0	22.9	26.1	27.4	25.8	20.2	13.6	7.8	6.4
	2%	WB	1.2	1.6	3.0	9.2	14.7	19.3	20.8	19.2	15.4	10.7	6.1	3.4
		MCDB	1.4	2.0	5.4	13.4	18.8	24.1	25.5	23.6	17.8	12.0	6.7	3.7
	5%	WB	0.5	0.8	1.7	7.5	13.0	17.6	19.6	18.0	14.2	9.6	5.1	2.0
		MCDB	0.8	1.3	3.1	11.2	17.7	21.8	23.7	21.7	16.3	10.6	5.7	2.4
	10%	WB	-0.2	0.0	0.9	5.6	11.5	15.9	18.5	16.9	13.2	8.5	3.9	1.0
		MCDB	0.1	0.5	2.0	8.7	16.2	20.1	22.2	20.0	15.0	9.4	4.4	1.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.1	6.8	7.7	9.0	8.5	8.1	7.7	6.3	4.5	4.0	4.6
		MCDBR	3.9	4.4	7.7	11.1	13.0	11.5	11.0	10.5	8.8	5.4	4.3	3.6
	5% WB	MCWBR	4.0	4.2	5.3	6.8	6.7	5.9	5.5	5.7	5.4	4.0	4.0	3.6
		MCDWR	3.8	4.4	6.3	9.9	11.2	10.1	9.4	9.1	7.6	4.9	4.1	3.6
(40) Clear-Sky Solar Irradiance	taub		0.247	0.296	0.321	0.361	0.356	0.349	0.383	0.368	0.337	0.318	0.264	0.216
		taud	2.028	2.082	2.152	2.294	2.437	2.483	2.406	2.426	2.496	2.391	2.120	1.970
	Edn at Noon		427	643	777	825	866	878	832	814	777	642	406	279
		Edh at Noon	38	73	98	105	99	97	102	92	72	55	35	22
(44) All-Sky Solar Radiation	RadAvg		0.23	0.82	2.26	3.81	4.97	5.38	5.09	3.88	2.21	0.84	0.23	0.11
	RadStd		0.03	0.18	0.31	0.38	0.41	0.36	0.57	0.36	0.24	0.12	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.60	N/A	N/A	N/A	N/A	N/A	-237	-229	N/A
(47) Regional (6 neighbors)		+0.55	N/A	N/A	N/A	N/A	N/A	-222	N/A	N/A

Nomenclature: See separate page